

MECHANICAL LEGEND

SYMBOL	DESCRIPTION
	DUCTWORK
	VARIABLE SPEED SWITCH
	THERMOSTAT
	GRILLE ON WALL
	SINGLE/DOUBLE LIGHT TROFFER
	LINEAR GRILLE
	SQUARE / ROUND DIFFUSER
	RETURN/EXHAUST AIR GRILLE
	CONTROL WIRING
	DUCT TAKE-OFF
	OPEN-ENDED DUCT
	FIRE EXTINGUISHER
	FIRE HOSE CABINET
	SPRINKLER - PENDANT
	SPRINKLER - CONCEALED
	SPRINKLER - UPRIGHT
	CONNECT TO EXISTING
	EXHAUST FAN
	FAN POWER BOX
	VENT PIPE
	SANITARY PIPE (FLOOR BELOW)
	PUMPED SANITARY PIPE
	DOMESTIC COLD WATER
	DOMESTIC HOT WATER
	GATE VALVE
	ELBOW TOWARDS
	ELBOW AWAY
	CLEANOUT
	CAP OFF
	METER
	TYPE-SIZE-CFM
	1 ST DOOR UNDERCUT
	ACCESS PANEL

DRAWING LIST

DWG No.	DESCRIPTION
M-01	LEGEND, DRAWING LIST & SPECIFICATION
M-02	EQUIPMENT LIST & DETAILS
M-03	DEMOLITION HVAC PLAN
M-04	NEW HVAC PLAN
M-05	DEMOLITION PLUMBING & FIRE PROTECTION PLAN
M-06	NEW PLUMBING & FIRE PROTECTION PLANS

LINE TYPE LEGEND

	EXISTING WORK TO REMAIN
	NEW WORK
	WORK TO BE DEMOLISHED/REMOVED

GENERAL NOTES

- THE MECHANICAL CONTRACTOR SHALL VISIT THE SITE TO VERIFY EXISTING CONDITIONS PRIOR TO COMMENCEMENT OF WORK, ORDERING OF EQUIPMENT AND/OR FABRICATING MATERIALS. THIS SHALL BE DONE IN ORDER TO CONFIRM THAT EQUIPMENT AND SERVICES CAN BE INSTALLED AS SHOWN ON DRAWINGS AND THAT ADDITIONAL COSTS ARE INCLUDED IN BID TO FACILITATE INSTALLATION. IT IS THE CONTRACTORS RESPONSIBILITY TO NOTIFY THE ENGINEERS OF ANY DISCREPANCIES, OMISSIONS, AND INTERFERENCES.
- COORDINATE WITH GENERAL CONTRACTOR TO SCHEDULE THE MECHANICAL CONTROLS AND PROTECTIVE WORKS, AS SPECIFIED ON PLANS. REFER TO, AND COMPLY WITH THE GENERAL CONTRACTOR'S PHASING PLAN.
- PROVIDE ALL REQUIRED CUTTING AND PATCHING OF EXISTING CEILINGS AND WALLS TO FACILITATE THE INSTALLATION OF THE MECHANICAL SERVICES OUTLINED FOR THIS SCOPE OF WORK.
- THIS CONTRACTOR SHALL PROVIDE ALL NECESSARY BALANCING DEVICES TO ACHIEVE PERFORMANCE SPECIFIED ON PLAN. NOT ALL AIR BALANCING DEVICE IS SHOWN ON PLAN FOR CLARITY.
- COORDINATE ALL SPRINKLER BRANCH PIPING WITH ALL NEW AND EXISTING EQUIPMENT AND SERVICES. RELOCATE SPRINKLER LINES TO FACILITATE THE INSTALLATION OF SERVICES AS REQUIRED.
- WHERE DUCT/PIPE IS REMOVED THROUGH FIRE-RATED WALL OR FLOOR ASSEMBLIES, PROVIDE ULC-CERTIFIED FIRE STOPPING SYSTEMS THAT ARE TESTED IN ACCORDANCE WITH CAN-515 AND INSTALL ALL FIRE STOPPING SYSTEM IN STRICT ACCORDANCE WITH THE MANUFACTURER'S INSTALLATION INSTRUCTION. STANDARD OF ACCEPTANCE: 3M, HILTI.
- RECORD AND REPORT ALL BASE BUILDING H.V.A.C. DEFICIENCIES TO LANDLORD/CONSULTANT.
- REFER TO SPECIFICATION FOR ALL DUCT INSULATION/LINING REQUIREMENTS.
- WHERE EXISTING DUCT/PIPE INSULATION IS DAMAGED DURING MECHANICAL DEMOLITION, PROVIDE NEW INSULATION TO MAINTAIN INTEGRITY OF THE EXISTING INSULATION.

GENERAL REQUIREMENTS:

- BASE BUILDING STANDARDS AND SPECIFICATIONS SHALL BE THE BASIS FOR THIS CONSTRUCTION.
- CONFORM TO THE OWNERS/LANDLORD'S TENANT DESIGN CRITERIA AND CONSTRUCTION MANUAL AND TENANT LEASE AGREEMENT.
- IDENTIFY ALL EQUIPMENT, DUCTS, VALVES, PIPES, ETC. TO BASE BUILDING STANDARDS.
- COMPLY WITH THE GENERAL CONTRACTOR'S CONSTRUCTION SCHEDULE.
- CONTACT LAWRENCE@LAWGICENGINEERING.COM PRIOR TO REINSTALLATION OF CEILINGS TO PERFORM A FINAL INSPECTION.
- DRAWINGS ARE TO BE CONSIDERED DIAGRAMMATICAL ONLY AND DO NOT SHOW ALL THE STRUCTURAL AND CONSTRUCTION DETAILS. VISIT THE SITE BEFORE SUBMITTING BID TO DETERMINE EXISTING CONDITIONS AND THE EXTENT TO WHICH THEY WILL AFFECT THE WORK UNDER THIS CONTRACT. VERIFY ALL SPACES IN WHICH EQUIPMENT, DUCTWORK, AND PIPING ARE TO BE LOCATED. FAILURE TO DO SO WILL NOT RELIEVE THIS CONTRACTOR FROM THE COST OF ADDITIONAL WORK.
- BEFORE ANY EQUIPMENT IS ROUGHED IN, DETERMINE ITS EXACT LOCATION FROM THE DRAWINGS. IF IT IS NOT SHOWN ON THE DRAWINGS, VERIFY EXACT LOCATION ON SITE. CONCEAL ALL SERVICES IN WALLS, CEILING SPACE, AND FLOOR SPACE UNLESS OTHERWISE STATED. LOCATIONS OF NEW AND EXISTING SERVICES ARE APPROXIMATE ONLY.
- OBTAIN PERMITS FROM THE MUNICIPALITY AND UTILITIES TO COMPLETE YOUR WORK. UPON COMPLETION OF YOUR WORK, SUPPLY AND TURN OVER TO THE MECHANICAL AND MUNICIPAL INSPECTION CERTIFICATES FROM GOVERNING AUTHORITIES TO CERTIFY THAT THE INSTALLED WORK CONFORMS TO THE RULES AND REGULATIONS OF GOVERNING AUTHORITIES. PAY FOR (INCLUDING ALL PROVINCIAL AND FEDERAL SALES TAXES) ALL PERMITS, FORMS, AND INSPECTIONS REQUIRED.
- SUPPLY ALL LABOUR, MATERIALS, TOOLS, SERVICES, EQUIPMENT, TRANSPORTATION, AND TESTING REQUIRED FOR THE SUPPLY AND INSTALLATION TO COMPLETE THE WORK SHOWN ON THE DRAWINGS AND SPECIFIED PRIOR.
- PROVIDE ALL CUTTING, PATCHING, FLASHING WORK, AND CLEAN-UP OF FLOORS, WALLS, AND CEILINGS REQUIRED FOR THE WORK.
- ALL MATERIALS IN CEILING SPACE SHALL CONFORM TO ONTARIO BUILDING CODE SECTION #6.4.3.
- SUPPLY ALL ACCESS DOOR/PANELS IN CEILINGS OR WALLS WHERE REQUIRED FOR ACCESS TO MECHANICAL EQUIPMENT THAT REQUIRES MAINTENANCE, BALANCING DAMPERS, FIRE DAMPERS, VALVES, FILTERS, FAN COILS, VAV BOX CONTROLS, ETC.) FOR INSTALLATION BY GENERAL TRADES. CONTRACTOR SHALL MINIMIZE QUANTITY OF ACCESS PANELS/DOORS BY GROUPING TOGETHER BALANCING DAMPERS AND VALVES. COORDINATE WITH ALL TRADES TO MINIMIZE ACCESS REQUIREMENTS. EQUIPMENT REQUIRING ACCESS SHALL BE CLEARLY IDENTIFIED BY PROJECT MANAGER AND BASE BUILDING OPERATOR PRIOR TO DRYWALL CEILING/WALL INSTALLATION. ACCESS DOOR/PANELS SHALL BE MILCOR OR LEHAGE OR EQUIVALENT, AND MUST BE COMPATIBLE WITH CEILING/WALL TYPE AND FINISH. FINISH SHALL SUIT ARCHITECT/INTERIOR DESIGNER'S REQUIREMENTS. PROVIDE 1-1/2 HOUR FIRE RATED DOOR/PANEL WHERE REQUIRED. COORDINATE ACCESS DOOR/PANEL LOCATIONS AND SIZES WITH OWNER/LANDLORD, BASE BUILDING OPERATOR, ARCHITECT/INTERIOR DESIGNER, AND PROJECT MANAGER.
- COORDINATE WITH ALL TRADES INSTALLING EQUIPMENT WHICH MAY AFFECT THE MECHANICAL WORK AND ARRANGE WATER RELATION WITH EQUIPMENT AND THE WORKING CONTRACTOR.
- OBTAIN AND PAY FOR ONE (1) SET OF WHITE PRINTS AND AUTOCAD FILES. OBTAIN AND PAY FOR CAD DRAWING DISK (\$40.00 PER DRAWING). WORK BEGINS ON SITE. CLEARLY AND ACCURATELY MARK-UP ALL CHANGES AND DEVIATIONS FROM THE ROUTING OF PIPING AND DUCTWORK AND LOCATIONS OF EQUIPMENT SHOWN ON THE CONTRACT DRAWINGS. CHANGES AND DEVIATIONS INCLUDE THOSE MADE BY ADDENDA, CHANGE ORDERS, AND SITE INSTRUCTIONS. MAINTAIN THE "AS-BUILT" WHITE PRINTS AT THE SITE FOR PERIODIC INSPECTION BY THE MECHANICAL CONSULTANT THROUGHOUT THE DURATION OF THE WORK. PAY PARTICULAR ATTENTION TO ACCURATELY DIMENSIONING THE LOCATION OF ALL CONCEALED SERVICES TERMINATED FOR FUTURE, ALL BURIED WORK AND SERVICES, AND CONCEALED WORK, BEFORE APPLYING FOR A CERTIFICATE OF COMPLETION. UPDATE AUTOCAD DISK SET(S) IN ACCORDANCE WITH THE MARKED UP "AS-BUILT" WHITE PRINTS. SUBMIT THE "AS-BUILT" SITE DRAWING WHITE PRINTS, WHITE PRINTS PRODUCED FROM THE DISK SET(S), AND DRAWING DISK SET(S) TO THE MECHANICAL CONSULTANT FOR REVIEW. UPON COMPLETION OF THE WORK, SUBMIT THE COMPLETED "AS-BUILT" DRAWINGS, TRANSPARENTS, AUTOCAD FILES, AND BUILDING INSPECTION DEPARTMENTS' FINAL CERTIFICATE OF APPROVAL TO THE MECHANICAL CONSULTANT AND OWNER/LANDLORD. "AS-BUILT" DRAWINGS SHALL CONTAIN THE CONTRACTOR'S NAME AND DATE.
- ARRANGE AND SCHEDULE ALL WORK IN A MANNER WHICH WILL NOT INTERFERE WITH NORMAL OPERATION OF THE EXISTING BUILDING, ALL SHUTDOWNS OF ANY PORTION OF EXISTING BASE BUILDING SYSTEMS SHALL BE PERFORMED BY THE OWNERS/LANDLORD'S BUILDING OPERATIONS STAFF AND/OR COORDINATED WITH THE OWNER/LANDLORD FOR TIME AND DURATION OF INTERRUPTIONS.
- ALL WORK AND EQUIPMENT PROVIDED SHALL BE IN COMPLIANCE WITH GOVERNING AUTHORITIES AND APPLICABLE CODES. WHERE GOVERNING AND MUNICIPAL STANDARDS AND SPECIFICATIONS ARE IN CONFLICT, THE MORE STRINGENT REQUIREMENT WILL APPLY.
- ENSURE MANUFACTURER'S INSTALLATION RECOMMENDATIONS AND INSTRUCTIONS ARE FOLLOWED UNLESS OTHERWISE NOTED HEREIN OR ON THE DRAWINGS. UNLESS SUCH INSTRUCTIONS AND RECOMMENDATIONS CONTRADICT GOVERNING CODES AND REGULATIONS.
- WHERE STANDARDS OF THE WORK ARE SPECIFIED OR IMPLIED AND THE WORK DOES NOT COMPLY WITH THE PERFORMANCE SPECIFIED OR IMPLIED, SUCH DEFICIENCY SHALL BE CORRECTED AS DIRECTED BY THE MECHANICAL CONSULTANT. TESTING AND RETESTING TO VERIFY THE WORK SHALL BE DONE AT THE CONTRACTOR'S EXPENSE. ANY CHARGES FOR THE OWNERS/LANDLORD'S STAFF, THE MECHANICAL CONSULTANT OR OTHER PERSONNEL RELATED TO SUCH RETESTING, SHALL ALSO BE AT THE CONTRACTOR'S EXPENSE.
- INCLUDE COST OF PREMIUM TIME IN TENDER PRICE FOR WORK DURING NIGHTS, WEEKENDS, OR OTHER TIME OUTSIDE NORMAL WORKING HOURS NECESSARY TO MAINTAIN ALL MECHANICAL SERVICES IN OPERATION. NOTE THAT OWNER/LANDLORD WILL CONTINUE TO OPERATE DURING NORMAL HOURS. THEREFORE, THE CONTRACTOR WILL BE RESTRICTED TO IRREGULAR HOURS. ALL WORK IN ADJACENT TENANT'S SPACE AND IN CEILING SPACE BELOW SHALL BE DONE AFTER NORMAL OFFICE HOURS AND COORDINATED WITH PROJECT MANAGER, OWNER/LANDLORD, AND ADJACENT TENANT.
- PROVIDE FIVE (5) COPIES OF PROPER SHOP DRAWINGS OF ALL SPECIFIED PRODUCTS AND SUBMIT FOR APPROVAL TO THE ARCHITECT/INTERIOR DESIGNER AND MECHANICAL CONSULTANT, FOLLOWING REVIEW AND APPROVAL BY CONTRACTOR (PROVIDE REVIEW STAMP). EQUIVALENT EQUIPMENT MAY BE PROPOSED PRIOR TO COMMENCEMENT OF WORK, PROVIDING THE QUALITY AND PERFORMANCE CHARACTERISTICS OF THE EQUIPMENT MEET THE SPECIFIED PRODUCTS. THE USE OF EQUIVALENT EQUIPMENT IS SUBJECT TO THE APPROVAL OF THE MECHANICAL CONSULTANT AND ARCHITECT/INTERIOR DESIGNER, AND ON SATISFACTORY SUBMISSION OF DETAILED SHOP DRAWINGS. ALLOW ONE (1) WEEK FOR EACH REVIEW. INCLUDE ONE (1) SET OF APPROVED SHOP DRAWINGS WITH OPERATION AND MAINTENANCE MANUAL (SEE BELOW).
- RETURN ALL REMOVED EQUIPMENT (EXHAUST FANS, DOMESTIC WATER HEATERS, ETC.) TO THE OWNER/LANDLORD OR DISPOSE OF EQUIPMENT AS REQUESTED BY THE OWNER/LANDLORD.
- IN ALL AREAS REQUIRING TRENCHING OR CORE DRILLING THROUGH EXISTING FLOOR SLAB FOR PLUMBING SERVICES, ALLOW FOR ALL NECESSARY RADIOGRAPHY TO LOCATE HIDDEN ELECTRICAL SERVICES, STRUCTURAL REINFORCING, ETC., AND INCLUDE ALL COSTS IN TENDER PRICE. COORDINATE THIS WORK WITH THE OWNER/LANDLORD AND/OR TENANT COORDINATOR REGARDING SCHEDULING, AND ADHERE TO THE OWNERS/LANDLORD'S REQUIREMENTS. SUBMIT CORE DRILLING PLAN TO BASE BUILDING STRUCTURAL ENGINEER FOR APPROVAL PRIOR TO COMMENCEMENT OF WORK.
- PROVIDE THE OWNER/LANDLORD A WRITTEN WARRANTY OF MINIMUM ONE (1) YEAR FOR THE COMPLETE MECHANICAL INSTALLATION FROM DATE OF ACCEPTANCE, INCLUDING ALL LABOUR, MATERIALS, AND EQUIPMENT IN THIS CONTRACT. REPAIR AND/OR REPLACE DEFECTS WHICH APPEAR IN YOUR WORK WITHIN THE WARRANTY. ORDINARY WEAR AND TEAR AND WILLFUL DAMAGE BY OR CARELESSNESS OF THE OWNERS/LANDLORD'S STAFF OR AGENTS EXCEPTED. WITHOUT ADDITIONAL EXPENSE TO THE OWNER/LANDLORD, WHERE SUCH DEFECTS OCCUR, BE RESPONSIBLE FOR ALL COSTS INCURRED IN MAKING DEMOLITION AND REPAIR WORK, REPAIR OR REPLACEMENT OF BUILDING FINISHES, OTHER MATERIALS, OR DAMAGE TO OTHER EQUIPMENT CAUSED BY SUCH DEFECTS, OR BY SUBSEQUENT REPLACEMENT AND REPAIRS.
- PROVIDE THREE (3) OPERATING AND MAINTENANCE MANUALS CONTAINING APPROVED SHOP DRAWINGS, AIR AND WATER BALANCING REPORTS, EQUIPMENT DATA SHEETS, WRITTEN DATA RECORDS, ALL STARTERS, CONTACTORS, MAINTENANCE PROCEDURES, TEST REPORTS, AND CONTACT LIST OF CONTRACTORS AND SUPPLIERS (WITH PHONE NUMBERS) TO MECHANICAL CONSULTANT FOR REVIEW. MANUALS SHALL BE IN A THREE-RING BINDER SEPARATED WITH DIVIDERS IN APPROPRIATE SECTIONS. MAKE ALL CORRECTIONS REQUESTED BY MECHANICAL CONSULTANT AND RESUBMIT FOR REVIEW.
- CHANGE NOTICE QUOTATIONS FOR EXTRA OR DELETED WORK SHALL BE SUBMITTED COMPLETE WITH ITEMIZED COST BREAKDOWN OF LABOUR AND MATERIALS. FAILURE TO PROVIDE WILL RESULT IN REJECTION. ALL MECHANICAL CHANGE NOTICES SHALL BE PRICED IN ACCORDANCE WITH MECHANICAL CONTRACTORS ASSOCIATION AND "ALL PRICES" LESS DISCOUNT FOR LABOUR AND MATERIAL COST.
- TEMPORARY 1" THICK FILTERS SHALL BE PROVIDED AT ALL BASE BUILDING RETURN AIR OPENINGS WHICH REMAIN OPERATIONAL DURING CONSTRUCTION, AND SHALL BE REPLACED WEEKLY. REMOVE UPON CONSTRUCTION COMPLETION. UPON COMPLETION OF WORK, INFORM BUILDING MAINTENANCE THAT ALL HVAC EQUIPMENT (NEW AND EXISTING) SHOULD HAVE FILTERS REPLACED.
- REMOVE ALL ITEMS PLANNED FOR DEMOLITION INCLUDING ANCILLARY SUPPLIES AND COMPONENTS. REMOVE MECHANICAL COMPONENTS AND ABANDONED COMPONENTS BACK TO SOURCE, CAP ALL DUCTS AND DRAINAGE PIPING, AND CAP AND VALVE ALL PIPING AT CORE RISER.
- PROVIDE INDEPENDENT SUPPORT FOR ALL COMPONENTS OF THE INSTALLATION.
- PROVIDE START-UP SERVICES AS PER THE MANUFACTURER'S RECOMMENDATION FOR THE NEW/RELOCATED EQUIPMENT SPECIFIED.
- PROVIDE ELECTRICAL DEVICES WITH WIRING, STARTERS, DISCONNECT, ETC. VERIFY AND COORDINATE VOLTAGE AND PHASE WITH THE ELECTRICAL CONTRACTOR PRIOR TO ORDERING OF EQUIPMENT. ALL STARTERS, CONTACTORS, RELAYS, ETC. SHALL BE SUPPLIED BY THE MECHANICAL CONTRACTOR AND INSTALLED BY THE ELECTRICAL CONTRACTOR. ALL POWER WIRING (LINE VOLTAGE) SHALL BE INSTALLED BY THE ELECTRICAL CONTRACTOR. ALL CONTROL WIRING (LOW VOLTAGE) SHALL BE BY THE MECHANICAL CONTRACTOR.
- DOMESTIC WATER SYSTEMS:**
- ALL DOMESTIC WATER PIPING SHALL BE TYPE "K" COPPER CERTIFIED TO ASTM B42, WITH CAST BRASS OR WROUGHT COPPER FITTINGS. ALL JOINTS SHALL BE MADE USING 95% TIN-ANTIMONY SOLDER. PRESSURE TEST ALL LINES IN ACCORDANCE TO LOCAL CODES BEFORE APPLYING TO SYSTEM.
- DOMESTIC WATER SOFT COPPER PIPING, WHERE NOTED ON DRAWINGS, SHALL BE TYPE "L" SOFT COPPER CERTIFIED TO ASTM B88 "SEAMLESS COPPER WATER TUBE" AND SUPPLIED IN A CONTINUOUS COIL OF PROPER LENGTH WITH NO JOINTS (IF POSSIBLE). WHERE REQUIRED, JOINTS SHALL BE COMPRESSION TYPE FLARED JOINT COUPLINGS TO ANSII B16.26.
- ALL SANITARY DRAINS AND MAIN VENT STACKS SHALL BE CAST IRON (DWV) TO CANCSA B70, COMPLETE WITH NJ JOINTS. BRANCH VENTS MAY BE COPPER TUBING, DWV SANITARY DRAINS UNDER B70 (75mm) AND CONDENSATE DRAINS MAY BE HARD TEMPERED COPPER DRAINAGE TUBE (DWV) TO ASTM B308, WITH SOLDER-JOINT DRAINAGE FITTINGS TO ASTM B16.23. PROVIDE TYPE "L" COPPER DRAIN PIPING FOR URINAL DRAIN PIPING FROM URINAL DRAIN OUTLET TO CAST IRON DRAIN STACK. (DWV) PIPES ARE NOT PERMITTED FOR URINAL DRAIN PIPING.
- VERIFY EXISTING LOCATIONS AND INVERT ELEVATIONS FOR SANITARY DRAINS ON SITE PRIOR TO COMMENCEMENT OF WORK.
- UNLESS OTHERWISE NOTED, SLOPE HORIZONTAL DRAINAGE PIPING B70 (75mm) AND SMALLER AT 2% SLOPE, AND PIPING LARGER THAN B70 (75mm) AT 1% SLOPE.
- PROVIDE CLEANOUTS SUITABLE IN ALL RESPECTS FOR THE INTENDED APPLICATION WHERE SHOWN ON THE DRAWINGS OR WHEN REQUIRED BY CODES. CLEANOUTS IN PIPING B70 (75mm) AND SMALLER SHALL BE SAME SIZE AS PIPE. OTHER CLEANOUTS SHALL BE A MINIMUM 4" (100mm) diameter.
- PROVIDE ISOLATING GATE VALVES ON MAIN AND/OR BRANCH LINES, AND AT ALL EQUIPMENT OR FIXTURES SERVED

- WITH HOT AND COLD WATER LINES. PROVIDE BACKFLOW PREVENTORS TO EQUIPMENT CONNECTIONS. ALL VALVES SHALL BE SUITABLE FOR THE OPERATING PRESSURE OF THE SYSTEM IN WHICH THEY ARE INSTALLED. MAKE AND MODEL SHALL BE AS PER BASE BUILDING STANDARDS AND SPECIFICATIONS, UNLESS NOTED OTHERWISE. INCREASED DIMENSIONS ACCORDINGLY.
- PROVIDE D-ELECTRIC COUPLINGS/UNIONS WHERE COUPLING PIPING CONNECTIONS TO FERROUS METAL AND PLUMBING EQUIPMENT SUCH AS STEEL STORAGE TANKS, PRVS, AND/OR STEEL, BLACK IRON, CAST IRON, OR GALVANIZED IRON PIPING.
- CHECK AND VERIFY LOCATION OF EXISTING MECHANICAL AND ELECTRICAL INTERFERENCES IN CEILING SPACE OF FLOOR BELOW INCLUDING STRUCTURAL FLOOR SLAB IN ALL AREAS REQUIRING CORE DRILLING AND/OR CUTTING OF FLOOR SLAB.
- FINAL LOCATION OF ALL NEW PLUMBING FIXTURES SHALL BE COORDINATED ON SITE WITH ALL TRADES. REFER TO ARCHITECTURAL DRAWINGS AND DETAILS FOR EXACT LOCATION OF PLUMBING FIXTURES. ALL PLUMBING FIXTURES SHALL BE PIPED COMPLETE WITH ALL NECESSARY APPURTENANCES, SUCH AS VENTS, SANITARY, HOT AND COLD WATER CONNECTIONS, ETC.
- INSTALL ALL COMPONENTS IN STRICT ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
- INSTALL SHOCK ARRESTORS ON HOT AND COLD WATER PIPING SERVING FIXTURES OR EQUIPMENT EQUIPPED WITH QUICK CLOSING VALVES.
- PLUMBING FIXTURES INCLUDING DOMESTIC HOT WATER HEATERS SHALL BE NEW, OF FIRST QUALITY, IN PERFECT CONDITION, AND INSTALLED IN BEST WORKMANLIKE MANNER. VERIFY PLUMBING FIXTURE QUANTITIES AND LOCATIONS WITH ARCHITECT/INTERIOR DESIGNER'S DRAWINGS. RELOCATION OF EXISTING DOMESTIC WATER HEATERS/STANKS IS STRICTLY NOT PERMITTED.
- PROVIDE TRAP SEAL PRIMER FOR ALL NEW HUB DRAINS.
- EXPOSED PIPING AND FITTINGS WITHIN WASHROOMS SHALL BE CHROME PLATED. PROVIDE CHROME PLATED ESCALATOR AND PASSING THROUGH FINISHING THROUGH FINISHING.
- PROVIDE INDEPENDENT SUPPORTS EVERY 6'-0" (1800mm) MINIMUM FOR 1/2" (25mm) OR LESS COPPER PIPING, AND EVERY 8'-0" (2400mm) OR OTHER REMOVABLE MARKINGS AT 1'-0" (300mm) OR LESS. PROVIDE INDEPENDENT SUPPORTS EVERY 8'-0" (2400mm) MINIMUM FOR ALL CAST IRON PIPING. REFER TO BASE BUILDING TENANT DESIGN MANUAL, AND USE WHICHEVER REQUIREMENT IS MORE STRINGENT.
- REMOVE ALL ABANDONED WATER SUPPLY PIPES, DRAIN LINES, AND VENT LINES BACK TO THE CORE RISER AND PROPERLY CAP AND/OR VALVE.
- PROVIDE SLEEVES FOR ALL PIPING PENETRATING WALLS AND FLOORS. ENSURE ALL PENETRATIONS THROUGH FIRE RATED ASSEMBLIES ARE PROPERLY SEALED WITH AN APPROVED COMPOUND.
- PLUMBING INSULATION:
 - ALL INSULATION AND INSTALLATION SHALL BE IN ACCORDANCE TO ASHRAE GUIDELINES.
 - ALL ABOVE GRADE PIPING IS TO BE INSULATED WITH FIBREGLASS INSULATION TO PREVENT CONDENSATION OR TO RETAIN HEAT FOR ENERGY EFFICIENCY. INSULATION SYSTEM TO BE SCHULLER MICRO-LOK OR EQUAL, WITH THE FOLLOWING SIZES:
 - 1" - SANITARY DRAINS (ANY SIZE)
 - 1" - DOMESTIC COLD WATER (ANY SIZE)
 - 1" - DOMESTIC HOT WATER (2" OR SMALLER)
 - 1/2" - DOMESTIC HOT WATER (2" OR LARGER)
 - 1" - DOMESTIC HOT WATER REGIC. (ANY SIZE)
 - 1/2" - DOMESTIC WATER (HOT OR COLD) BRANCHES
 - MAKE GOOD INSULATION AT NEW/EXISTING INSULATION WORK INTERFACES.
- HVAC DUCTWORK/PIPPING SYSTEM:**
- ALL NEW MATERIALS AND EQUIPMENT SHALL BE PROVIDED AND INSTALLED IN ACCORDANCE WITH BASE BUILDING STANDARDS.
- ALL DUCTWORK AND HANGERS SHALL BE FABRICATED IN ACCORDANCE WITH THE LATEST ASHRAE AND SMACNA RECOMMENDATIONS AND STANDARDS.
- SEAL ALL JOINTS TO SMACNA SEAL CLASS A.
- RECTANGULAR OR SQUARE GALVANIZED STEEL DUCTWORK SHALL BE PRIME LOCK FORMING QUALITY TO ASTM A525M. SATIN COATED FINISH ON DUCTWORK TO BE PAINTED. G60 COATING ON ALL OTHER DUCTWORK, WITH METAL GAUGES IN ACCORDANCE WITH PUBLISHED SMACNA "HVAC DUCT CONSTRUCTION STANDARDS METAL AND INSULATION" TO SUIT THE DUCT CONFIGURATION AND CLASSIFICATION.
- PROVIDE FLEXIBLE CONNECTORS BETWEEN ALL FANS AND ADJACENT DUCTWORK CONSISTING OF:
 - 1" - DOMESTIC HOT WATER (2" OR SMALLER)
 - 1/2" - DOMESTIC HOT WATER (2" OR LARGER)
 - 1" - DOMESTIC HOT WATER REGIC. (ANY SIZE)
 - 1/2" - DOMESTIC WATER (HOT OR COLD) BRANCHES
- FIBREGLASS FLEXIBLE WITH ELASTOMER COATING, DURO DYNE "DUROLON" OR EQUIVALENT.
- FLEXIBLE DUCTWORK SHALL BE ALUMINUM SPIRAL, WOUND CLASS ONE ULC APPROVED, FLEXMASTER "TRIPLE LOCK" OR EQUIVALENT. SECURE TO RIGID DUCT AND AT NECKS OF DIFFUSERS USING GEAR CLAMPS AND SEAL AIR TIGHT WITH DUCT SEALER. FLEXIBLE DUCTS SERVING DIFFUSERS SHALL BE INSTALLED AS ONE CONTINUOUS PIECE OF DUCT. IF JOINTS ARE REQUIRED, THEY SHALL BE ROUND RIGID DUCT. FLEXIBLE DUCTS SHALL BE OF DIAMETER EQUAL TO DIFFUSER NECK SIZE OR AS NOTED OTHERWISE.
- MAKE GOOD ALL EXISTING INSULATION WHEN CONNECTING TO EXISTING SERVICES.
- AIR TRANSFER DUCTS SHALL BE PROVIDED WHEREVER REQUIRED TO ENSURE ADEQUATE RETURN AIR AND/OR SMOKE EXHAUST AIR IN CEILING SPACE BACK TO RETURN AIR GENERAL EXHAUST AIR, AND SMOKE EXHAUST AIR OPENINGS. WHERE THERE IS A MULTI-TENANT CORRIDOR, ENSURE THAT TRANSFER AIR DUCTS ARE NOT INSTALLED IN CORRIDOR WALLS - REFERENCE MECHANICAL CONSULTANT OF ANY EXISTING TRANSFER AIR DUCTS FOUND IN CORRIDOR WALLS, AND ADJACENT TENANT.
- RESPONSIBILITY TO COORDINATE AND CONFIRM AIR TRANSFER PROVISIONS WITH THE GENERAL TRADES.
- AIR TRANSFER OPENINGS INCLUDING WITHOUT DUCT (BAFFLE OPENINGS AND/OR DOOR UNDERCUTS) SHALL BE THIS CONTRACTOR'S RESPONSIBILITY. COORDINATE AND CONFIRM PROVISIONS WITH GENERAL TRADES.
- NEW SUPPLY AIR DIFFUSERS/REGISTERS AND RETURN/EXHAUST AIR GRILLES SHALL MATCH BASE BUILDING OR BE OF TYPE AS INDICATED ON DRAWINGS. COORDINATE FINAL LOCATION WITH LATEST ARCHITECTURAL REFLECTED CEILING PLANS. RELOCATE AND REUSE EXISTING DIFFUSERS AND GRILLES WHERE POSSIBLE OR WHERE INDICATED ON PLANS.
- PROVIDE BALANCING DAMPERS FOR ALL NEW AND EXISTING DUCT BRANCHES ON EXISTING DUCTWORK WITH SUITABLE MEANS OF CEILING ACCESS FOR BALANCING, AND VOLUME DAMPERS FOR ALL NEW SUPPLY AIR DIFFUSERS AND REGISTERS.
- SPIN ON DAMPERS SHALL BE OF TYPE (NAILOR MODEL 1890 - MANUAL BALANCING DAMPER).
- TEST, BALANCE, AND ADJUST AIR SYSTEMS TO OBTAIN THE DESIGN AIR QUANTITIES (+10%), MARK THE FINAL BALANCE POSITION ON ALL BALANCING DAMPERS AND ADJUSTABLE AIR TURNING DEVICES.
- SUBMIT THREE (3) AIR TESTING AND BALANCING REPORTS TO THE MECHANICAL CONSULTANT AND TENANT COORDINATOR. INDICATE ALL TEST RESULTS INCLUDING MAXIMUM AND MINIMUM AIR FLOW OF EACH DIFFUSER, SPACE THERMOSTAT SETTING, CLOSEST AND FURTHEST OUTLET SUPPLY AIR TEMPERATURES, AND ROOM TEMPERATURES. PROVIDE A WRITTEN REPORT OF THE TESTING. THIS WORK SHALL BE PERFORMED BY THE OWNERS/LANDLORD'S TESTING AND BALANCING CONTRACTOR OR BY A CONTRACTOR APPROVED BY THE OWNER/LANDLORD, AND COST SHALL BE INCLUDED UNDER THIS CONTRACT. BALANCING CONTRACTOR SHALL BE A MEMBER OF ASAC OR NEBC. SUBMIT THE TESTING AND BALANCING REPORTS TO THE MECHANICAL CONSULTANT AND THE OWNER/LANDLORD INDICATING TERMINAL DESIGN AND MEASURED FLOW RATES. PROVIDE SIX (6) ADDITIONAL HOURS OF BALANCING WORK, TO BE PERFORMED AFTER THE TENANT HAS MOVED IN FOR "COMFORT BALANCING."
- PRIOR TO INSTALLATION, OBTAIN ARCHITECT/INTERIOR DESIGNER'S APPROVAL ON LOCATION OF ALL REGISTERS, DIFFUSERS, AIR TROFFERS, THERMOSTATS, ACCESS PANELS, ETC.
- VERIFY ALL FITTINGS AND FLANGES ARE NOT PERMITTED.
- ELECTRIC FITTINGS SHALL BE USED WHERE REQUIRED.
- ALL FITTINGS ARE TO BE ASME CLASS 300 RATED.
- CHILLED WATER PIPING AND HEATING WATER PIPING (WHERE SHOWN) SHALL BE SCHEDULE 40 BLACK STEEL TO ASTM A53, PIPING 2" (50mm) AND SMALLER SHALL BE THREADED TO ANSI B31.1, INSULATE ALL HEATING AND CHILLED WATER PIPING, FITTINGS, AND VALVES WITH 1" (25mm) THICK FIBREGLASS INSULATION (5.5 LBS DENSITY) AND COVER INSULATION WITH FACTORY APPLIED FIRE RESISTIVE FIBREGLASS REINFORCED KRAFT PAPER AND ALUMINUM FOIL VAPOUR BARRIER. SEAL INSULATION VAPOUR TIGHT. DO NOT USE STAPLES.
- PRESSURE TEST ALL NEW PIPING TO 1.5 TIMES THE WORKING PRESSURE FOR A PERIOD OF EIGHT (8) HOURS BEFORE APPLYING INSULATION. INFORM THE MECHANICAL CONSULTANT BEFORE COMMENCEMENT OF TEST. SUBMIT SUCCESSFUL TEST REPORT TO MECHANICAL CONSULTANT.
- ALL SHUT-OFF VALVES AND BALANCING VALVES SHALL BE RATED FOR 300 PSI (1380 KPA) WORKING PRESSURE.
- STRAINERS SHALL BE RATED FOR 250 PSI (1725 KPA) WORKING PRESSURE, AND ALL PIPE FITTINGS SHALL BE RATED FOR 300 PSI (2070 KPA) WORKING PRESSURE. STRAINERS SHALL BE BRONZE BODY TYPE WITH SCREWED CONNECTIONS, STAINLESS STEEL SCREENS WITH 1/16" (1.6mm) PERFORATIONS, AND CAPABLE OF SYSTEM PRESSURE OF 125 PSI (860 KPA).
- INSTALL AUTOMATIC AIR VENTS AT ALL HIGH POINTS OF NEW PIPING SYSTEM. ALL VALVES SHALL BE AS PER BASE BUILDING AND SPECIFICATIONS UNLESS OTHERWISE STATED.
- PROVIDE D-ELECTRIC COUPLINGS FOR CONNECTION OF DISSIMILAR PIPING MATERIALS.
- PROVIDE CIRCUIT BALANCING VALVES TO BALANCE WATER FLOW. CIRCUIT BALANCING VALVES SHALL BE ARMSTRONG MODEL CBV1 - V PATTERN STYLE, ALL METAL, WITH SOLDEDED OR SCREWED CONNECTIONS, BUILT-IN DRAIN CONNECTION, PROTECTIVE CAPS, AND INTEGRAL VALVE INSULATION. EACH VALVE SHALL BE PROVIDED WITH VERNIER TYPE HAND/HEEL SETTINGS FOR PRECISION FLOW BALANCING. POSITIVE SHUT OFF VALVE WITH NO DRP SEAT AND PLUG TYPE STEM WITH TEFLON DISC. TAMPER PROOF HIDDEN MEMORY, POSITIVE SHUT OFF METERING VALVES WITH CONNECTIONS FOR PORTABLE METER.
- SELECT CIRCUIT BALANCING VALVES TO GIVE A PRESSURE DROP AT 100% OPEN BETWEEN 1 FT HEAD (3.0 KPA) AND 7 FT HEAD (240 KPA) AT 100% OPEN. PROVIDE THE FULL TUBE TO ANSI B31.1 IN THE CIRCUIT NEAR MINIMUM PRESSURE DROP AND THOSE LOCATED NEAR THE PUMPS AT HIGHER PRESSURE DROPS.
- TEST, BALANCE AND ADJUST WATER SYSTEMS TO OBTAIN THE DESIGN WATER FLOW RATES (45%).
- WATER BALANCING SHALL BE PROVIDED FOR ALL CHILLED/HEATING WATER SYSTEMS AND SHALL INCLUDE A WRITTEN REPORT INDICATING TEMPERATURE, FLOW RATES, OPERATING PRESSURES, AND PRESSURE DIFFERENTIAL BETWEEN THE SUPPLY AND RETURN AT EACH PIECE OF EQUIPMENT, WHEN REQUIRED. BALANCING SHALL BE COMPLETED BY OWNER/LANDLORD'S TESTING AND BALANCING CONTRACTOR AND PAID FOR UNDER THIS CONTRACT. SUBMIT FOUR (4) WATER TESTING AND BALANCING REPORTS TO THE MECHANICAL CONSULTANT AND TENANT COORDINATOR.
- PROVIDE SAFETY AND RELIEF VALVES FOR ALL CLOSED WATER SYSTEMS. PIPE RELIEF TO NEAREST FLOOR DRAIN. PROVIDE WAITS 174A VALVES INSTALLED AT 150 PSI (1035 KPA) AT 210°F (99°C) ASTM RATED, CAST IRON BODY BRONZE DISC AND SEAT, STEEL SPINDLE ASSEMBLY, CARBON STEEL SPRING.
- DUCTWORK INSULATION:
 - ALL INSULATION AND INSTALLATION SHALL BE IN ACCORDANCE TO ASHRAE GUIDELINES.
 - PROVIDE 1" (25MM) THICK ACOUSTIC INSULATION FOR NEW SUPPLY, RETURN OR EXHAUST DUCTWORK

10'-0" FROM AIR HANDLING UNIT. RIGID SUPPLY AIR DUCTWORK SHALL BE THERMALLY INSULATED.

- ALL DUCTWORK DIMENSIONS ARE CLEAR INSIDE DIMENSIONS. WHERE ACOUSTIC INSULATION IS INSTALLED, INCREASED DIMENSIONS ACCORDINGLY.
- INSULATE ALL NEW SUPPLY AIR RIGID DUCTWORK UP TO 30" WIDE (750MM) WIDE OR DIAMETER WITH 1" (25MM) THICK FLEXIBLE GLASS FIBRE DUCT INSULATION, 1.5 LB/FT³ (118.42 KG/M³) DENSITY WITH FIRE RESISTIVE GLASS FIBRE REINFORCED KRAFT PAPER AND ALUMINUM FOIL VAPOUR BARRIER.
- ACOUSTIC LINING MATERIAL SHALL BE JOHNS-MANVILLE "LINACOUSTIC PERMACOTE HP", 1" (25 MM) THICK ACOUSTIC LINING MATERIAL MEETING NFPA 90A AND ASTM C1071, G21 AND G22 REQUIREMENTS, NOT SUPPORTING MICROBIAL GROWTH AND FLAME SPREAD AND SMOKE DEVELOPED FIRE HAZARD RATINGS OF CANA S102, CONSISTING OF A BONDED FIBREGLASS MAT COATED ON THE INSIDE (AIR SIDE) FACE WITH A BLACK FIRE RESISTANT COATING."
- MAKE GOOD INSULATION AT NEW/EXISTING INSULATION WORK INTERFACES.
- TERMINAL UNITS:**
- VARIABLE VOLUME TERMINALS
- PROVIDE CONTROLS SYSTEMS TRAINING FOR OWNER/LANDLORD WHEN SYSTEM HAS BEEN COMPLETED AND VERIFIED AS PER SPECIFICATIONS.
- PROVIDE SINGLE DUCT, VARIABLE AIR VOLUME TERMINALS OF THE SIZES AND CAPACITIES SHOWN IN THE DRAWINGS.
- TERMINALS SHALL BE CERTIFIED UNDER THE ARI STANDARD 880-94 CERTIFICATION PROGRAM AND CARRY THE ARI SEAL.
- THE TERMINAL CASING SHALL BE MINIMUM 0.8MM GALVANIZED STEEL, INTERNALLY LINED WITH NON-POROUS SEALED LINER WHICH COMPLIES WITH UL 181 AND NFPA 90A. INSULATION SHALL BE 24 KG/M³ DENSITY.
- THE DAMPER SHALL HAVE HEAVY GAUGE STEEL WITH SHAFT ROTATING IN BRONZE OIL-SELF-LUBRICATING BEARINGS. NYLON BEARINGS ARE NOT ACCEPTABLE. SHAFT SHALL BE CLEARLY MARKED ON THE END TO INDICATE DAMPER POSITION. STOCKED OR OTHER REMOVABLE MARKINGS ARE NOT ACCEPTABLE. THE DAMPER SHALL INCORPORATE A MECHANICAL STOP TO PREVENT OVERTSTROKING AND A SYNTHETIC SEAL TO LIMIT CLOSURE LEAKAGE.
- MAXIMUM CASING AIR LEAKAGE SHALL BE TWO (2) PERCENT AT 747 PA.
- ACTUATORS SHALL BE MOUNTED EXTERNALLY FOR SERVICE ACCESS. TERMINALS WITH INTERNAL ACTUATOR MOUNTING OR LINKAGE CONNECTION MUST INCLUDE GASKETED ACCESS PANEL. REMOVABLE WITHOUT DISTURBING DUCTWORK.
- AT AN INLET VELOCITY OF 10.2 M/S, THE DIFFERENTIAL STATIC PRESSURE REQUIRED TO OPERATE ANY TERMINAL SIZE SHALL NOT EXCEED 6MM W.G. FOR THE BASIC TERMINAL.
- SOUND RATINGS FOR THE TERMINAL SHALL NOT EXCEED 35 NC AT 75MM W.G. STATIC PRESSURE. SOUND PERFORMANCE SHALL BE ARI CERTIFIED.
- CONTROLS SYSTEM:**
- THE TERMINAL SUPPLIER SHALL PROVIDE THE AIRFLOW SENSOR, AND CONTROL TRANSFORMER.
- AIRFLOW SENSOR SHALL BE DESIGNED TO PROVIDE A DIFFERENTIAL PRESSURE SIGNAL, WHICH IS AMPLIFIED OVER THE FULL CAPACITY RANGE OF THE TERMINAL. PRESSURE MEASURING TAPS SHALL BE PROVIDED EXTERNAL TO THE UNIT.
- PROVIDE ALL NECESSARY INTERNAL CONTROL TUBING, WIRING, AND MOUNTING BRACKETS FOR A COMPLETE OPERATING UNIT.
- ALL CONTROL COMPONENTS SHALL BE MOUNTED INSIDE A PROTECTIVE METAL SHROUD PROVIDED BY THE BOX MANUFACTURER.
- PRIMARY AIR DELIVERY SHALL BE PRESSURE INDEPENDENT. ROOM TEMPERATURE CONTROL SHALL OPERATE SATISFACTORILY AT PRIMARY SUPPLY DUCT STATIC PRESSURES RANGING FROM 240 TO 1245 PA. MAXIMUM AND MINIMUM BOX VOLUMES SHALL BE FACTORY SET AND CALIBRATED. SETTINGS SHALL BE FIELD ADJUSTABLE.
- INSTALL VAV, FPAV, AND CAV BOXES IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
- EACH BOX SHALL BE CLEARLY MARKED WITH AN IDENTIFICATION LABEL LISTING THE BOX TAG NUMBER, AND MINIMUM AND MAXIMUM AIR SETTINGS.
- SUSPEND BOXES FROM SLAB USING THREADED ROD HANGERS AND ANGLE IRON TRAPEZE HANGERS. HANGERS FOR FPAV BOXES SHALL INCORPORATE NEOPRENE ISOLATORS.
- ALL CONTROL WIRING SHALL BE CARRIED OUT BY DIV 16; POWER WIRING SHALL BE BY DIV 16. THE CONTROL SYSTEM SHALL BE SUPPLIED AND INSTALLED COMPLETE IN ACCORDANCE WITH THE SPECIFICATIONS. DEMONSTRATE TO THE MECHANICAL CONSULTANT ON COMPLETION OF WORK.
- ALL CONTROLS WORK SHALL BE PERFORMED BY THE OWNERS/LANDLORD'S CONTROLS CONTRACTOR OR A CONTRACTOR APPROVED BY THE OWNER/LANDLORD.
- PROVIDE ALL CONTROLS, EMT CONDUIT, FITTINGS, AND WIRING INCLUDING APPURTENANCES NECESSARY FOR COMPLETE AND OPERATING CONTROL SYSTEMS. HARD WIRE ALL ELECTRICAL CONTROL DEVICES INTO THE ASSOCIATED SYSTEM MAGNETIC STARTER. PROVIDE POWER TO CONTROL PANEL FROM THE NEAREST NORMAL POWER ELECTRICAL DISTRIBUTION PANEL.
- MOUNTING HEIGHT OF THERMOSTATS SHALL BE 4'-14" ON CENTRE OF CONTROL HEIGHT FROM FINISHED FLOOR. COORDINATE LOCATION WITH ARCHITECT/INTERIOR DESIGNER. DO NOT INSTALL IN VICINITY OF ELECTRICAL LIGHTING DIMMERS.
- CLEAN AND RECALIBRATE ALL EXISTING THERMOSTATS UPON COMPLETION OF CONSTRUCTION. SUBMIT REPORT THAT THIS WORK HAS BEEN COMPLETED.
- DUCTWORK/PIPPING INSULATION:**
- UNLESS OTHERWISE NOTED, ALL INSULATION MATERIALS MUST MEET REQUIREMENTS OF NFPA 90A AND MUST HAVE A FIRE HAZARD RATINGS OF NOT MORE THAN 25 FOR FLAME SPREAD AND 50 FOR SMOKE DEVELOPED WHEN TESTED IN ACCORDANCE WITH CANULC S102.
- FIBREGLASS PIPE INSULATION FOR DOMESTIC HOT AND COLD WATER PIPING TO 2" (50mm) DIAMETER:
 - JOHNS-MANVILLE "MICRO LOK", RIGID, MOULDED, SECTIONAL FIBREGLASS PIPE INSULATION WITH A FACTORY APPLIED "AP 1 PLUS" VAPOUR BARRIER JACKET WITH PRESSURE SENSITIVE, LONGITUDINAL LAP SEALING SYSTEM. SECURE THE LAP OF THE INSULATION JACKET IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS. BUTT TOGETHER ADJOINING SECTIONS OF THE INSULATION AND COVER THE BUTT JOINTS WITH STRIPS OF VAPOUR BARRIER JACKET MATERIAL, SECURED IN PLACE WITH ADHESIVE. INSULATE ELBOWS, VALVES AND SIMILAR FITTINGS WITH FACTORY FORMED FIBREGLASS INSULATION OF A THICKNESS AND INSULATION VALUE EQUAL TO THAT OF SECTION INSULATION. LAMINATE IN PLACE WITH ADHESIVE AND SECURE WITH TWINE. APPLY A SMOOT COAT OF INSULATING GEMENT. WRAK THE FITTING WITH CLOTH TAPE OVERLAPPED 50% AND SET IN ADHESIVE, AND APPLY A COAT OF FINISHING CEMENT WITH A SMOOTH FINISH OR, INSTEAD OF THE CEMENT AND CLOTH TAPE, APPLY A SURE FIT SYSTEM "SMOKE LESS" 5050 RATED PVC FITTING COVERS.
 - AT EACH PIPE HANGER LOCATION (PIPE SUPPORTED WITHOUT ROLLER HANGER), PROVIDE A 1' (50mm) LENGTH OF PITTSBURGH CORNING "FOAMGLAS" MOULDED, SECTIONAL, CELLULAR GLASS PIPE INSULATION IN LIEU OF SPECIFIED FIBREGLASS INSULATION, WHERE INSULATED PIPE IS "COLD PIPE". WRAK INSULATION WITH DOUBLE THICKNESS OF ADJACENT INSULATION. LAMINATE IN PLACE WITH ADHESIVE AND SECURE WITH TWINE. APPLY A SMOOT COAT OF INSULATING GEMENT. WRAK THE FITTING WITH CLOTH TAPE OVERLAPPED 50% AND SET IN ADHESIVE, AND APPLY A COAT OF FINISHING CEMENT WITH A SMOOTH FINISH OR, INSTEAD OF THE CEMENT AND CLOTH TAPE, APPLY A SURE FIT SYSTEM "SMOKE LESS" 5050 RATED PVC FITTING COVERS.
- FILL PIPE SADDLE VOID WITH FIBREGLASS INSULATION.
- PROVIDE 1/2" (15mm) THICK INSULATION TO DOMESTIC COLD WATER AND ALL CONDENSATE DRAINS. PROVIDE 1" (25mm) INSULATION TO 2" (50mm) OR LESS DOMESTIC HOT WATER, 2" (50mm) OR LESS BUILDING HOT WATER, 6" (150mm) OR LESS CHILLED WATER, 4" (100mm) OR LESS CONDENSER WATER, ALL HORIZONTAL. SANITARY AND STORM DRAINS, AND 3" (75mm) OR LESS HEAT TRACED PIPE. PROVIDE 1-1/2" INSULATION TO 2-1/2" (65mm) OR LARGER DOMESTIC HOT WATER, 2-1/2" (65mm) OR LARGER BUILDING HOT WATER, 5" (125mm) OR LARGER CONDENSER WATER, AND 4" (100mm) OR LARGER HEAT TRACED PIPE.
- PROVIDE ALL LABOUR, MATERIALS, PRODUCTS, EQUIPMENT, AND SERVICES TO SUPPLY AND INSTALL THERMAL INSULATION, VAPOUR BARRIERS, AND FINISHES FOR MECHANICAL WORK AS INDICATED ON THE DRAWINGS AND SPECIFIED IN THIS SECTION OF THE SPECIFICATIONS.
- MAINTAIN AMBIENT TEMPERATURES AND CONDITIONS REQUIRED BY MANUFACTURERS OF ADHESIVES, MASTICS, AND INSULATING CEMENTS.
- INSULATION MATERIALS MUST BE MANUFACTURED AT FACILITIES CERTIFIED AND REGISTERED WITH AN APPROVED REGISTRAR TO CONFORM TO ISO 9000 QUALITY STANDARD.
- ALL INSULATION PERTAINING TO DIVISION 15 SHALL BE CARRIED OUT BY ONE FIRM SPECIALIZING IN INSULATION WORK. DO NOT MIX SIMILAR PRODUCTS OF MULTIPLE MANUFACTURERS.
- PROVIDE INSULATION MATERIALS WITH A MINIMUM THERMAL CONDUCTIVITY OF 0.24 BTU·IN/(HR. SQ·FT·°F) AT 100°F (38°C) MEAN TEMPERATURE.
- ON HOT PIPING APPLICATIONS, HOLD INSULATION IN PLACE WITH FLARE TYPE STAPLES (OUTWARD CLIPPING).
- ON COLD PIPING APPLICATIONS, APPLY VAPOUR BARRIER JACKET OVER INSULATION AND SEAL LONGITUDINAL AND CIRCUMFERENTIAL LAPS WITH CHILDERS CP82 OR BAKELITE 230-39 ADHESIVE. SEAL ALL PIPE TERMINATIONS, INCLUDING FITTINGS, WALP PENETRATIONS, AND PIPE SUPPORTS WITH VAPOUR BARRIER MASTIC. FOR CHILLED WATER SYSTEMS, PROVIDE VAPOUR SEAL PIPE TERMINATIONS EVERY FOUR (4) JOINT SECTIONS.
- APPLY PIPE INSULATION OVER 1-1/2" (40mm) THICKNESS IN TWO (2) LAYERS WITH JOINTS STAGGERED.
- INSULATE FITTINGS WITH FABRICATED MITERED OR PREFORMED SECTIONS OF SPECIFIED INSULATION.
- INSULATE OVER FLANGES AND MECHANICAL COUPLERS WITH SPECIFIED INSULATION AND THICKNESS, SIZED TO SUIT FLANGE DIAMETERS. FIT SPACES BETWEEN INSULATION AND ADJOINING PIPE INSULATION WITH SIMILAR MATERIAL.
- INSULATE VALVES AND INLINE COMPONENTS WITH FLEXIBLE INSULATION (3.4 LB/SCU FT DENSITY) COMPRESSED NOT TO EXCEED ORIGINAL THICKNESS. BUILD UP TO SPECIFIED THICKNESS WITH APPROVED ASBESTOS FREE FINISHING CEMENT.
- DO NOT INSULATE TERMINAL UNIT AUTOMATIC CONTROL VALVES INSTALLED IN HOT PIPING. DO NOT INSULATE TERMINAL UNIT AUTOMATIC CONTROL VALVES WHICH ARE INSTALLED IN COLD PIPING AND WHICH ARE LOCATED OVER CONDENSATE DRAIN PANS.
- UNDER ALL HANGERS USED ON CHILLED WATER AND DOMESTIC COLD WATER, PROVIDE AN INSERT BETWEEN SUPPORT SHIELD AND PIPING FOR PIPING 1-1/2" (40mm) OR LARGER.
- DUCTWORK INSULATION:
 - JOHNS-MANVILLE "800 SERIES SPIN GLASS FSK", RIGID FIBREGLASS INSULATION WITH A FACTORY APPLIED, FIBREGLASS REINFORCED, FOIL AND FLAME RETARDANT KRAFT PAPER (FSK) VAPOUR BARRIER FACING.
 - JOHNS-MANVILLE "MICROLOTE FSK", FLEXIBLE BLANKET TYPE FIBREGLASS INSULATION WITH A FACTORY APPLIED, FIBREGLASS REINFORCED, FOIL AND FLAME RETARDANT KRAFT PAPER (FSK) VAPOUR BARRIER FACING. INSULATION FOR CASINGS, PLENUMS AND EXPOSED RECTANGULAR DUCTWORK SHALL BE RIGID BOARD TYPE. INSULATION FOR ROUND DUCTWORK AND CONCEALED RECTANGULAR DUCTWORK SHALL BE BLANKET TYPE.

- SECURELY BUTT TOGETHER ADJOINING SECTIONS OF INSULATION, SECURED IN PLACE WITH FULL COVERAGE OF ADHESIVE ON ALL SURFACES. IN ADDITION TO ADHESIVE, ON VERTICAL AND SUSPENDED SURFACES, SECURE INSULATION WITH PINS WELDED TO THE DUCTWORK ON 12" (300mm) TO 18" (450mm) CENTRES WITH THE INSULATION APPLIED OVERTOP OF PINS AND SECURED WITH CLIPS. ENSURE THAT THE INSULATION DOES NOT SAG OR BULGE.
- MAKE JOINTS IN VAPOUR BARRIER FACINGS WITH 3" (75mm) wide OVERLAPPING STRIPS OF INSULATION VAPOUR BARRIER FACINGS SECURED WITH A

M-02



CLIENT:



ENGINEER:



SEAL:



PROJECT ADDRESS:

SUITE 800,
70 YORK ST.,
TORONTO, ON
M5H 1J8

ISSUANCE/REVISION SCHEDULE

[illegible]

ISSUED FOR PERMIT

SCALE: $1/8" = 1'-0"$

ISSUED DATE: 12-02-24

RAWN BY: MA

HECKED BY: LL

SHEET NAME:

DEMOLITION HVAC PLAN

SHEET NO.:

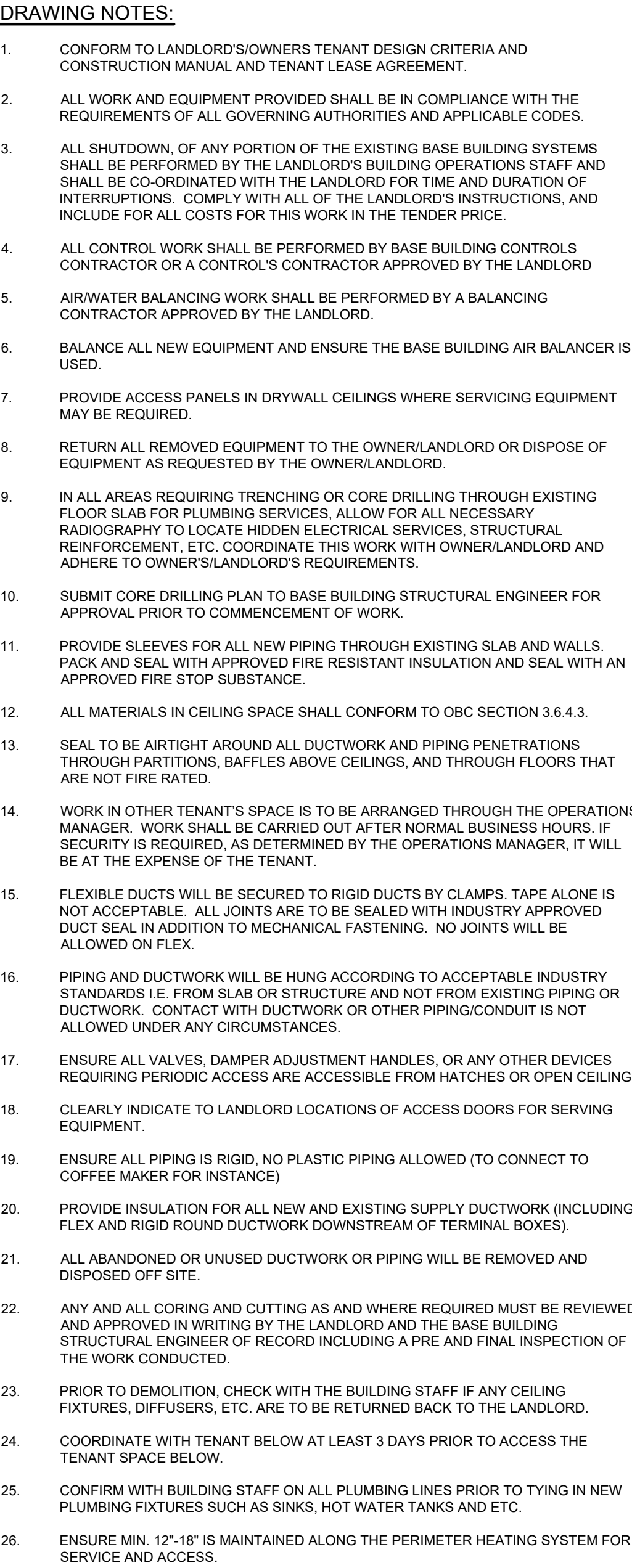
M-03

DRAWING NOTES:

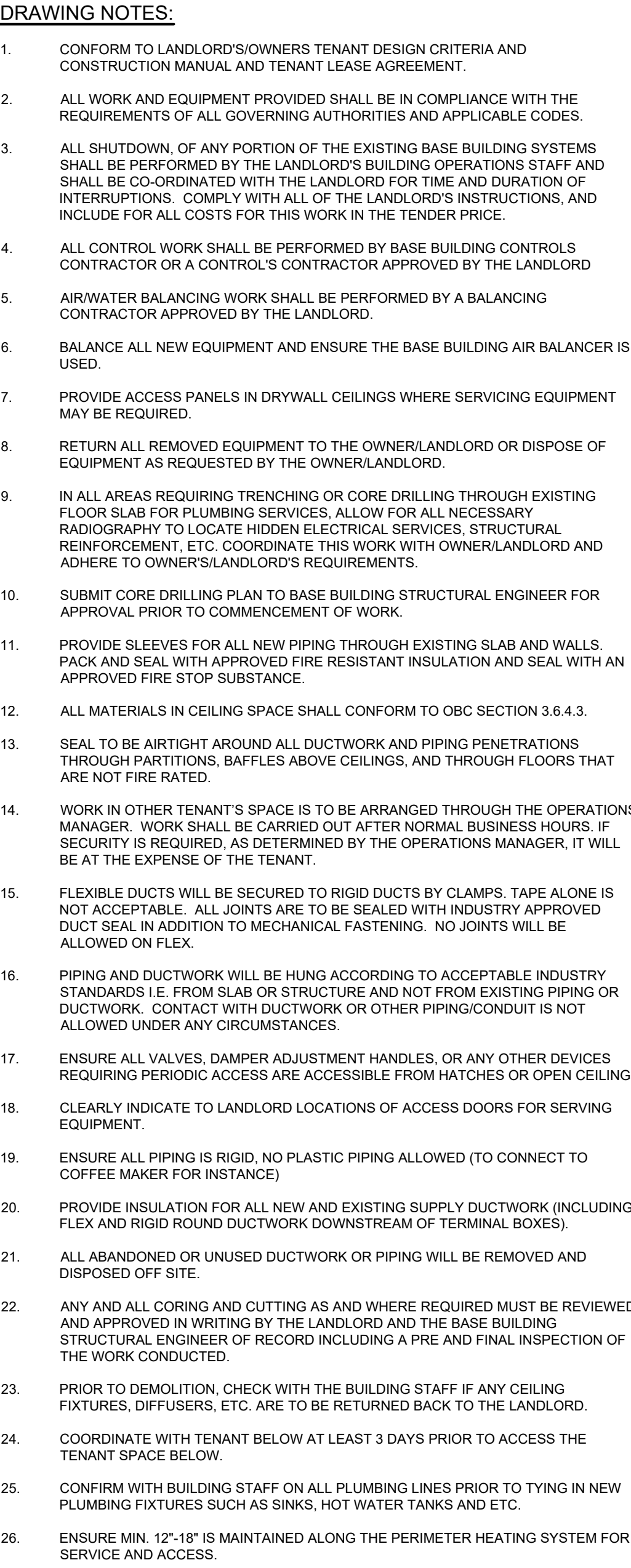
1. CONFORM TO LANDLORD'S/OWNERS TENANT DESIGN CRITERIA AND CONSTRUCTION MANUAL AND TENANT LEASE AGREEMENT.
2. ALL WORK AND EQUIPMENT PROVIDED SHALL BE IN COMPLIANCE WITH THE REQUIREMENTS OF ALL GOVERNING AUTHORITIES AND APPLICABLE CODES.
3. ALL SHUTDOWN, OF ANY PORTION OF THE EXISTING BASE BUILDING SYSTEMS SHALL BE PERFORMED BY THE LANDLORD'S BUILDING OPERATIONS STAFF AND SHALL BE CO-ORDINATED WITH THE LANDLORD FOR TIME AND DURATION OF INTERRUPTIONS. COMPLY WITH ALL OF THE LANDLORD'S INSTRUCTIONS, AND INCLUDE FOR ALL COSTS FOR THIS WORK IN THE TENDER PRICE.
4. ALL CONTROL WORK SHALL BE PERFORMED BY BASE BUILDING CONTROLS CONTRACTOR OR A CONTROL'S CONTRACTOR APPROVED BY THE LANDLORD
5. AIR/WATER BALANCING WORK SHALL BE PERFORMED BY A BALANCING CONTRACTOR APPROVED BY THE LANDLORD.
6. BALANCE ALL NEW EQUIPMENT AND ENSURE THE BASE BUILDING AIR BALANCER IS USED.
7. PROVIDE ACCESS PANELS IN DRYWALL CEILINGS WHERE SERVICING EQUIPMENT MAY BE REQUIRED.
8. RETURN ALL REMOVED EQUIPMENT TO THE OWNER/LANDLORD OR DISPOSE OF EQUIPMENT AS REQUESTED BY THE OWNER/LANDLORD.
9. IN ALL AREAS REQUIRING TRENCHING OR CORE DRILLING THROUGH EXISTING FLOOR SLAB FOR PLUMBING SERVICES, ALLOW FOR ALL NECESSARY RADIOGRAPHY TO LOCATE HIDDEN ELECTRICAL SERVICES, STRUCTURAL REINFORCEMENT, ETC. COORDINATE THIS WORK WITH OWNER/LANDLORD AND ADHERE TO OWNER'S/LANDLORD'S REQUIREMENTS.
10. SUBMIT CORE DRILLING PLAN TO BASE BUILDING STRUCTURAL ENGINEER FOR APPROVAL PRIOR TO COMMENCEMENT OF WORK.
11. PROVIDE 9L GUEVES FOR ALL NEW PIPING THROUGH EXISTING SLAB AND WALLS. PACK AND SEAL WITH APPROVED FIRE RESISTANT INSULATION AND SEAL WITH AN APPROVED FIRE STOP SUBSTANCE.
12. ALL MATERIALS IN CEILING SPACE SHALL CONFORM TO OBC SECTION 3.6.4.3.
13. SEAL TO BE AIRTIGHT AROUND ALL DUCTWORK AND PIPING PENETRATIONS THROUGH PARTITIONS, Baffles ABOVE CEILINGS, AND THROUGH FLOORS THAT ARE NOT FIRE RATED.
14. WORK IN OTHER TENANT'S SPACE IS TO BE ARRANGED THROUGH THE OPERATIONS MANAGER. WORK SHALL BE CARRIED OUT AFTER NORMAL BUSINESS HOURS. IF SECURITY IS REQUIRED, AS DETERMINED BY THE OPERATIONS MANAGER, IT WILL BE AT THE EXPENSE OF THE TENANT.
15. FLEXIBLE DUCTS WILL BE SECURED TO RIGID DUCTS BY CLAMPS. TAPE ALONE IS NOT ACCEPTABLE. ALL JOINTS ARE TO BE SEALED WITH INDUSTRY APPROVED DUCT SEAL. IN ADDITION TO MECHANICAL FASTENING, NO JOINTS WILL BE ALLOWED ON FLEX.
16. PIPING AND DUCTWORK WILL BE HUNG ACCORDING TO ACCEPTABLE INDUSTRY STANDARDS I.E. FROM SLAB OR STRUCTURE AND NOT FROM EXISTING PIPING OR DUCTWORK. CONTACT WITH DUCTWORK OR OTHER PIPING/CONDUIT IS NOT ALLOWED UNDER ANY CIRCUMSTANCES.
17. ENSURE ALL VALVES, DAMPER ADJUSTMENT HANDLES, OR ANY OTHER DEVICES REQUIRING PERIODIC ACCESS ARE ACCESSIBLE FROM HATCHES OR OPEN CEILING.
18. CLEARLY INDICATE TO LANDLORD LOCATIONS OF ACCESS DOORS FOR SERVICING EQUIPMENT.
19. ENSURE ALL PIPING IS RIGID, NO PLASTIC PIPING ALLOWED (TO CONNECT TO COFFEE MAKER FOR INSTANCE)
20. PROVIDE INSULATION FOR ALL NEW AND EXISTING SUPPLY DUCTWORK (INCLUDING FLEX AND RIGID ROUND DUCTWORK DOWNSTREAM OF TERMINAL BOXES).
21. ALL ABANDONED OR UNUSED DUCTWORK OR PIPING WILL BE REMOVED AND DISPOSED OFF SITE.
22. ANY AND ALL CORING AND CUTTING AS AND WHERE REQUIRED MUST BE REVIEWED AND APPROVED IN WRITING BY THE LANDLORD AND THE BASE BUILDING STRUCTURAL ENGINEER OF RECORD INCLUDING A PRE AND FINAL INSPECTION OF THE WORK CONDUCTED.
23. PRIOR TO DEMOLITION, CHECK WITH THE BUILDING STAFF IF ANY CEILING FIXTURES, DIFFUSERS, ETC. ARE TO BE RETURNED BACK TO THE LANDLORD.
24. COORDINATE WITH TENANT BEFORE AT LEAST 3 DAYS PRIOR TO ACCESS THE TENANT SPACE BELOW.
25. CONFIRM WITH BUILDING STAFF ON ALL PLUMBING LINES PRIOR TO TYING IN NEW PLUMBING FIXTURES SUCH AS SINKS, HOT WATER TANKS AND ETC.
26. ENSURE MIN. 12" R-18 IS MAINTAINED ALONG THE PERIMETER HEATING SYSTEM FOR SERVICE AND ACCESS.

KEYED NOTES (REFER TO FLOOR PLAN) :

- 1 REMOVE EXISTING AIR GRILLES, DIFFUSERS AND TROFFERS AND ASSOCIATED BRANCH DUCTWORK AND CAP BACK TO MAIN WHERE POSSIBLE. RETAIN AND RETURN ALL GRILLES AND DIFFUSERS TO LANDLORD IN GOOD WORKING CONDITION. (TYPICAL)
- 2 EXISTING PERIMETER RADIATION UNITS TO REMAIN. UNITS SHALL BE TESTED FOR PROPER OPERATION PRIOR TO DEMOLITION. REPORT ALL DEFICIENCIES TO LANDLORD. (TYPICAL)
- 3 CONTRACTOR TO PERFORM AIR AUDIT OF EXISTING SPACE PRIOR TO STARTING ANY HVAC MODIFICATIONS. SUBMIT TO ENGINEER FOR REVIEW
- 4 SEAL ANY OPENINGS AND CHECK AND REPAIR ANY DAMAGED INSULATION ON MAIN DUCTS. (TYPICAL)
- 5 EXISTING TRANSFER AIR DUCT ON DEMISING WALL TO REMAIN AS IS. (TYPICAL OF 2)
- 6 EXISTING VAV TERMINAL UNITS TO REMAIN. UNITS SHALL BE TESTED FOR PROPER OPERATION PRIOR TO DEMOLITION. (TYPICAL)
- 7 REMOVE EXISTING THERMOSTATS SERVING VAV TERMINAL UNITS C/W ASSOCIATED CONTROL WIRING AND RELOCATE TO NEW LOCATION. REFER TO NEW PLANS FOR NEW LOCATIONS. (TYPICAL)
- 8 REMOVE BRANCHES AS SHOWN BACK TO TERMINAL UNIT OR MAIN DUCT TO ALLOW FOR NEW DUCTWORK. REFER TO NEW FLOOR PLAN FOR NEW DUCTWORK CONNECTIONS. (TYPICAL)
- 9 RELOCATE EXISTING VAV TERMINAL UNITS AS SHOWN. REFER TO NEW FLOOR PLAN FOR NEW TERMINAL UNIT LOCATIONS. REMOVE ASSOCIATED DUCTWORK AND CAP BACK TO MAIN DUCT. PROVIDE NEW VAV AS REQUIRED. (TYPICAL)
- 10 TEST ALL EXISTING THERMOSTATS AND CONTROLLERS PRIOR TO DEMOLITION. VERIFY OPERATION OF EXISTING THERMOSTATS PRIOR TO DEMOLITION AND REPORT ALL DEFICIENCIES TO LANDLORD. (TYPICAL)
- 11 EXISTING RETURN AIR DUCT TO MECHANICAL ROOM TO REMAIN.
- 12 EXISTING PERIMETER LINEAR DIFFUSERS TO REMAIN. (TYPICAL)
- 13 REPAIR/REPLACE ALL DAMAGED BRANCH DUCTS SERVING VAV TERMINAL UNITS AS REQUIRED. (TYPICAL)
- 14 PROVIDE AND INSTALL A SOUND ATTENUATOR DOWNSTREAM OF THE VAV TERMINAL UNIT. REFER TO M-02 FOR MORE DETAILS.
- 15 CAP BACK DUCT TO ADJACENT SUITE. COORDINATE WITH ADJACENT SUITE PRIOR TO STARTING WORK.



1. NEW 24"x24" SQUARE PLAQUE SUPPLY AIR DIFFUSER (6"Ø INLET, UNLESS OTHERWISE NOTED) c/w BALANCING DAMPER. (TYPICAL)
2. NEW 12"x6" ALUMINUM SIDE WALL GRILLE c/w INTEGRATED BALANCING DAMPER MOUNTED ON DUCTWORK AS SHOWN. (TYPICAL)
3. PROVIDE NEW 24"x12" CEILING MOUNTED EGGRATE RETURN AIR GRILLE IN ALL ROOMS WITH TBAR CEILING AS SHOWN. (TYPICAL)
4. BALANCE ALL AIR OUTLETS TO AIRFLOW QUANTITIES SHOWN. (TYPICAL)
5. NEW ACOUSTICALLY LINED TRANSFER AIR DUCT c/w FACE GRILLE AS SHOWN. SIZES SHOWN IS INTERNAL, FREE AREA DIMENSIONS. TRANSFER AIR DUCTS AND GRILLS TO BE INSTALLED NOT LOWER THAN 9" FROM THE FLOOR OR IF THIS HEIGHT IS NOT POSSIBLE, PLEASE CONFIRM WITH THE SITE SUPERVISOR FOR FINAL PLACEMENT. MINIMUM 5' LONG ACOUSTICALLY LINED DUCT. REFER TO EQUIPMENT LIST & DETAILS (02).
6. BUILDING OPERATOR SHALL ENSURE MIN. 15% OUTDOOR AIR IS SUPPLIED TO THE SPACE AND RETURN AIR IS ADEQUATE FOR CIRCULATION.
7. NEW TRANSFER AIR FACE GRILLE ON WALL AS SHOWN. (TYPICAL OF 2)
8. ALL NEW/RELOCATED THERMOSTATS SHALL BE VERIFIED ON SITE FOR PROPER OPERATIONS. COORDINATE WITH CONTRACTOR ON SITE FOR FINAL INSTALLATION OF THERMOSTATS. OBTAIN ALL REMOVED PERIMETER TERMINAL FROM LANDLORD. PERIMETER THERMOSTATS SHALL BE INTEGRATED WITH PERIMETER RADIATION UNITS TO PREVENT SIMULTANEOUS HEATING AND COOLING. (TYPICAL)
9. ALL EXISTING PERIMETER RADIATION UNITS SHALL BE CLEANED AND TESTED FOR PROPER OPERATION. REPORT ALL DEFICIENCIES TO LANDLORD. (TYPICAL)
10. ENSURE ALL EXISTING AND NEW/RELOCATED PERIMETER TERMINAL UNITS ARE TIED WITH RADIATION UNITS TO PREVENT SIMULTANEOUS HEATING AND COOLING. (TYPICAL)
11. NEW/RELOCATED TERMINAL UNIT c/w ASSOCIATED CONTROLS. ALL NEW TERMINAL UNITS SHALL BE MATCH BASE BUILDING AND TIED TO BUILDING BAS. (TYPICAL)
12. PROVIDE NEW TRANSFER AIR GRILLE ON BOTH FACE OF WALL AS SHOWN. (TYPICAL)
13. PROVIDE ACOUSTICAL Baffles INSIDE PERIMETER RADIATION UNIT TO ATTENUATE SOUND BETWEEN ADJACENT SPACES/ROOMS SEPARATED BY WALL AS SHOWN. (TYPICAL)



- ① EXISTING 100' FHC TO REMAIN.
- ② REMOVE EXISTING SPRINKLERS. CAP ALL BRANCHES BACK TO MAIN AND MAKE READY FOR NEW SPRINKLERS ON NEW PLAN. RETAIN AND RETURN ALL REMOVED SPRINKLERS TO LANDLORD. (TYPICAL)
- ③ EXISTING DCW PIPE TO REMAIN.

M-06



1. REMOVE EXISTING NORMAL AND EMERGENCY POWER LIGHT FIXTURES. REFER TO NEW PLANS. EXTEND EXISTING WIRING AS REQUIRED. (TYPICAL)
2. REMOVE EXISTING EXIT SIGNS AND REPLACE WITH NEW ONES. EXTEND EXISTING WIRING AS REQUIRED. (TYPICAL)
3. REMOVE AND RELOCATE EXISTING FIRE ALARM SPEAKERS TO SUIT NEW LAYOUT. EXTEND EXISTING WIRING AS REQUIRED. WORK TO BE COMPLETED BY APPROVED CONTRACTOR BY BASE BUILDING. (TYPICAL)
4. REMOVE ALL EXISTING RECEPTACLES AND DATA PORTS TO SUIT NEW LAYOUT. ALL ABANDONED OR UNUSED WIRING SHALL BE REMOVED AND CAPPED BACK TO MAIN. CONTRACTOR SHALL VISIT SITE PRIOR TO TENDER TO VERIFY EXISTING CONDITIONS. (TYPICAL)
5. REMOVE ALL EXISTING NETWORK CABLES AND EQUIPMENT THROUGHOUT SPACE AND SERVER ROOM.
6. REMOVE ALL EXISTING BX CABLES IN CEILING SPACE AND REDISTRIBUTE IN CONDUITS TIED TIGHT TO THE UNDERSIDE OF SLAB. (TYPICAL)
7. REMOVE ALL ABANDONED OR UNUSED RECEPTACLE OUTLETS AND ABANDONED WIRING MOUNTED ABOVE THE CEILING AND RUN WIRING BACK TO SOURCE. (TYPICAL)
8. REMOVE ALL EXISTING SERVER EQUIPMENT, RACKS, CABLES AND ACCESSORIES IN THE EXISTING SERVER ROOM AND CAP BACK TO SOURCE. REMOVE ALL EXISTING TELEPHONE LINES AND ETC. COORDINATE WITH LANDLORD PRIOR TO DEMOLITION.

clear_space

CLEARSPACE OFFICES INC.
CONTACT@CLEARSPACE.TO

CLIENT:



BENTALL GREEN OAK

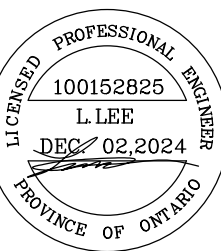
ENGINEER:



Lawgic
Engineering

LAWGIC ENGINEERING LTD.
INFO@LAWGICENGINEERING.COM

SEAL:



PROJECT ADDRESS:

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70 YORK ST.,
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ISSUANCE/REVISION SCHEDULE

[illegible]

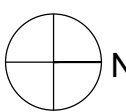
ISSUED FOR PERMIT

SCALE: 1/8" = 1'-0"

ISSUED DATE: 12-02-24

DRAWN BY: MA

CHECKED BY: LL



SHEET NAME:

EXISTING ELECTRICAL PLAN

SHEET NO.:

E-02

E-03